

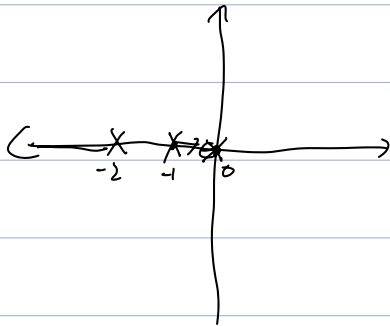
Root Locus.

$$\frac{C(s)}{R(s)} = \frac{K \cdot \frac{B(s)}{A(s)}}{1 + K \cdot \frac{B(s)}{A(s)}}$$

$$1 + K \frac{B(s)}{A(s)} = 0 \Rightarrow \frac{B(s)}{A(s)} = -\frac{1}{K}$$

e.g. $\frac{1}{s(s+1)(s+2)} = -\frac{1}{K} \Rightarrow s(s+1)(s+2) = -K$

$$s^3 + 3s^2 + 2s + K = 0$$

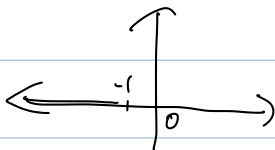


T2.

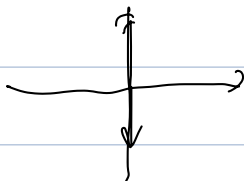
Q1, Close loop: $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$

Close loop pole: $1 + G(s) = 0$

(a) $1 + \frac{K}{s+1} = 0 \Rightarrow s = -1-K \quad (K > 0)$



(b) $1 + \frac{K}{s^2} = 0 \Rightarrow s = \pm \sqrt{K} = \pm \sqrt{K}j \quad (K > 0)$



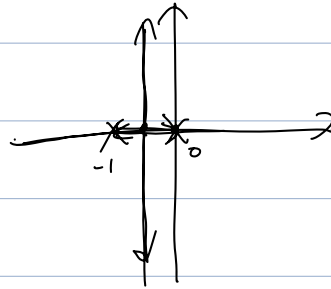
Asymptotes

(b). 2 poles: 0 and -1, $n=2$

0 zeros: $m=0$

$(n-m)=2$ asymptotes

$$\chi = \frac{0-1}{2} = -0.5$$



$$\frac{dG}{ds} = \frac{-K(2s+1)}{s^2(s+1)^2} = 0 \Rightarrow s = -\frac{1}{2} \quad 1 + \frac{K}{s(s+1)} = 0$$

$$\Rightarrow s^2 + s + K = 0$$

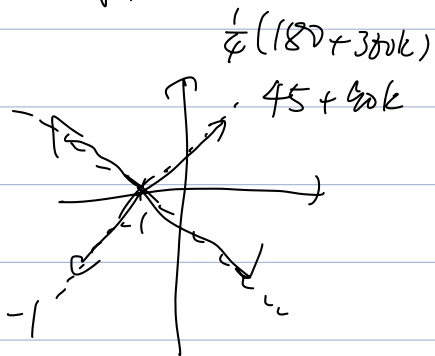
$$\Rightarrow s = -\frac{1}{2} \pm \sqrt{\frac{1}{4} - K}$$

(d): 4 poles at -1, $n=4$

0 zeros, $m=0$

There are $n-m=4$ asymptotes

$$\text{which centers at } \chi = \frac{-1-1-1-1}{4} = -1$$



$$\text{Double points: } \frac{dG_0}{ds} = -4(s+1)^{-5} = 0 \Rightarrow s = -1$$

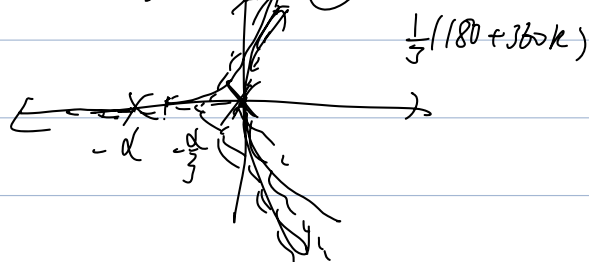
Q3. (a). 3 poles: two at 0, one at -d, $n=3$

0 zeros.

$$3 \text{ asymptotes: } \chi = \frac{0+0-d}{3} = -\frac{d}{3}$$

$$\text{Double points: } \frac{dG_0}{ds} = \frac{-(3s^2+2sd)}{s^4(s+d)^2} = \frac{-(3s+2d)}{s^3(s+d)^2} = 0$$

$$\Rightarrow s = -\frac{2}{3}d \text{ or } 0$$

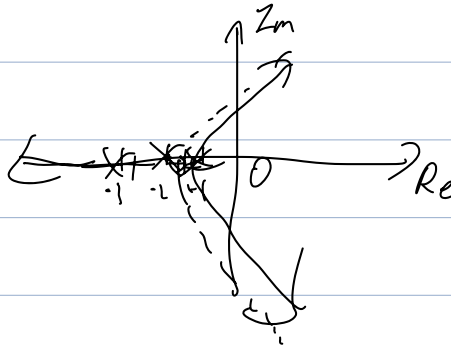


$$(b) \quad -1 \quad -2 \quad -3$$

$$\frac{k}{s^3 + 6s^2 + 11s + 6}$$

$$-(3s^2 + 12s + 11) = 0$$

$$s_1 = -2.58 \quad s_2 = -1.42$$



$$(c) \quad (i) \quad G(s) = \frac{K(1+5s)}{s^2(1+s)}$$

3 poles: two at 0, one at -1, $n: 3$

1 zero: $-\frac{1}{5}$, $m: 1$

$(n-m) = 2$ asymptotes

$$\frac{1}{2}(180^\circ + 360^\circ k)$$

$90^\circ + 180^\circ k$

center at $\sigma = \frac{(0+0-1) - (-\frac{1}{5})}{2} = -0.4$

double points: $\frac{dG(s)}{ds} = \frac{5s^2(1+s) - (1+5s)(2s+3s^2)}{s^4(1+s)^2} = 0$

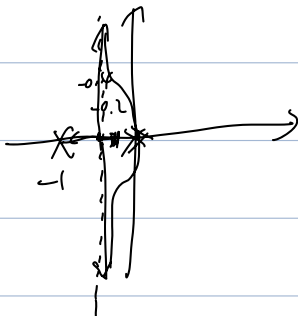
$$5s^2 + 5s^3 - 2s - 3s^2 - 10s^2 - 15s^3 = 0$$

$$-10s^3 - 8s^2 - 2s = 0$$

$$s(5s^2 + 4s + 1) = 0$$

$$0, \quad \frac{-4 \pm \sqrt{16-20}}{10} = \frac{-4 \pm 2j}{10}$$

$$0, \quad -0.4 \pm 0.2j$$



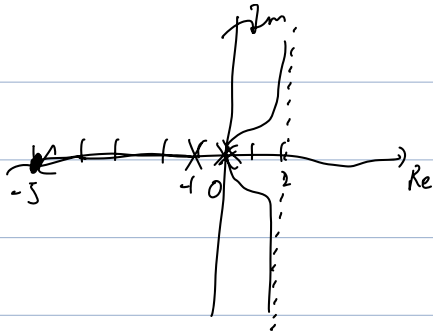
$$(ii) \quad G(s) = \frac{K(1+0.2s)}{s^2(1+s)}$$

3 poles: 0, 0, -1, $n: 3$, 3 branches.

1 zero: -5, $m: 1$

$n-m = 2$ asymptotes

centers at $\sigma = \frac{0+0-1 - (-5)}{2} = 2$



$$0.2(s^2 + s^3) - (1 + 0.2s)(2s + 3s^2) = 0$$

$$0.2s[s + s^2 - (5 + s)(2 + 3s)] = 0$$

$$0.2s[s + s^2 - 3s^2 - 17s - 10] = 0$$

$$-2s^2 - 16s - 10$$

$$s^2 + 8s + 5$$

$$\frac{-8 \pm \sqrt{64 - 20}}{2}$$